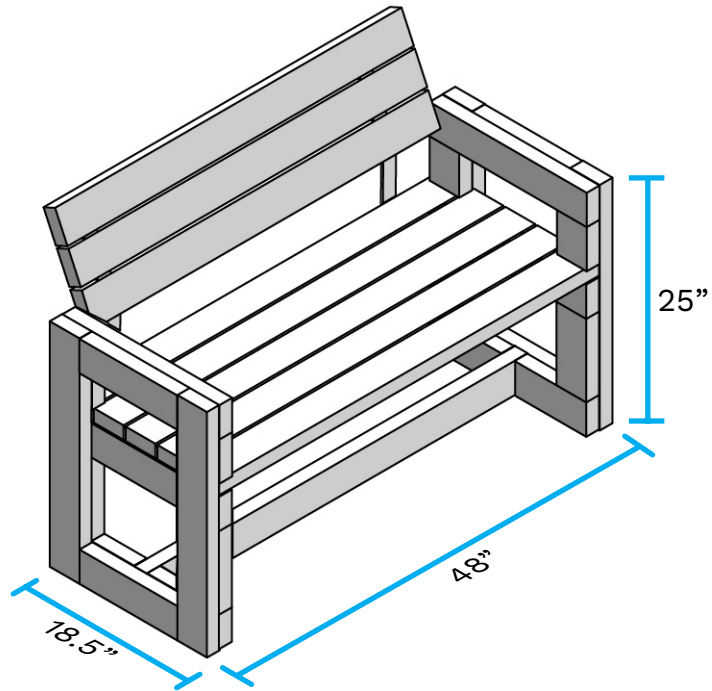




BENCH + BACK

ASSEMBLY INSTRUCTIONS

BEFORE YOU BEGIN

1. This document includes recommendations for do-it-yourself (DIY) construction of a **bench + back**. As a DIY project, the Owner, or the Owner's Contractor, is required to procure, fabricate, assemble, install, and maintain all project materials. These are recommendations only and any material, assembly, and construction must be tailored to specific project requirements.
2. These recommendations are based on common best practices associated with standard DIY home-improvement and/or landscape construction. These best practices may include working knowledge and experience with manual and power tools, hardware, fasteners, lumber, concrete footings, and associated techniques of assembly and construction. Prior working knowledge of these best practices is required so as to avoid harm or personal injury to self or others, or to property.
3. This document includes recommendations for the sequence and assembly of **bench + back** components. Locate your **bench + backs** in your plan you created in the 'Dream' workbook to find the proposed location and dimension of all **bench + backs**. This plan must be tailored to the specific conditions of the project site, including property lines, existing features- such as walls, trees, etc- and any other existing site conditions that may limit or otherwise influence the installation. Use your cost estimation spreadsheet to determine the quantity needed for all your **bench + backs**.
4. Before using any power equipment (such as drills, saws, augers, or other), review all manufacturer safety requirements. While handling or operating any power equipment, follow all manufacturer safety requirements. Do not allow others who have not reviewed and/or follow manufacturer safety requirements to handle or utilize power equipment.
5. The owner is solely responsible for any work required to secure, and comply with, all necessary building and safety codes, permits, and regulatory approvals.
6. These recommendations assume that the project site has been sufficiently prepared for construction. This includes, but is not limited to: identification of utilities that may interfere or otherwise impact installation; the removal of trash, hazards, unwanted structures; or other conditions that may inhibit safe construction.
7. Before construction, review the project site conditions to identify any potential limitations to assembly and installation. This includes determining advance notification by the Owner of utility companies so as to determine the location of all above and below ground utilities.
8. All persons associated with building and construction should sign a volunteer waiver.

LEGAL NOTICE:

The recommendations included herein are based on standard practices of fabrication and construction of garden furnishings. In no event shall Thomas Jefferson University (TJU), TJU's Lab For Social Innovation, Park-in-a-Truck, and/or its faculty, staff or affiliates be liable for any damages or injury to person or property damages arising out of the use of these recommendations. Thomas Jefferson University (TJU), TJU's Lab For Social Innovation, Park-in-a-Truck, or its faculty, staff and/or affiliates make no warranties, expressed or implied, and hereby disclaims and negates all other warranties, including without limitation, implied warranties or conditions of merchantability, fitness for a particular purpose, or non-infringement of intellectual property or other violation of rights. Further, Thomas Jefferson University (TJU), TJU's Lab For Social Innovation, Park-in-a-Truck, and/or its faculty, staff or affiliates do not warrant or make any representations concerning the accuracy, likely results, or reliability of the use of these recommendations. The use of these recommendations constitutes your acknowledgment of the aforementioned.

BENCH + BACK

ASSEMBLY INSTRUCTIONS

COMMON LUMBER SELECTIONS \

WESTERN RED CEDAR:

PRO: NATURALLY WEATHER/ROT RESISTANT LUMBER WITH AESTHETICALLY PLEASING RED TONES AND PLEASANT AROMAS, THAT IS EASY TO WORK WITH. NO PRESERVATION TREATMENTS ARE NECESSARY AS THE SURFACES WILL GRADUALLY DEVELOP A LOVELY GRAY PATINA. IT IS EASILY RECYCLABLE AT THE END OF ITS PRIMARY USE.

CON: COST, BE SURE TO SPECIFY 'KNOTTY' (STK) GRADE WHICH IS MUCH MORE AFFORDABLE THAN 'CLEAR' GRADE.

DOUGLAS FIR:

PRO: MODERATELY WEATHER/ROT RESISTANT AND STRONG LUMBER. IT IS ALSO EASY TO WORK WITH AT A LOWER COST THAN CEDAR. IT IS EASILY RECYCLABLE AT THE END OF ITS PRIMARY USE.

CON: IT WILL NOT STAND THE TEST OF TIME AS WELL AS CEDAR, BUT A GREAT CHOICE ESPECIALLY IF YOU PLAN TO PAINT IT.

PRESSURE TREATED:

PRO: WEATHER/ROT RESISTANT LUMBER AT A LOW COST.

CON: LUMBER IS OFTEN OF VARIABLE POOR QUALITY AND SATURATED WITH CHEMICAL TREATMENT MAKING IT DIFFICULT TO WORK WITH LEADING TO INCONSISTENT RESULTS. NEEDS 1-4 MONTHS OF AIR DRYING BEFORE PAINTING. NOT EASILY RECYCLABLE AND OFTEN ENDS UP IN LANDFILLS AT THE END OF ITS PRIMARY USE. NOT RECOMMENDED FOR USE IN AREAS WHERE VEGETABLES ARE GROWN.

GENERAL NOTES:

1. WHEN ORDERING LARGE QUANTITIES, ORDER AT LEAST 5% MORE LUMBER. TO ALLOW FOR SOME VARIABLE LUMBER QUALITY AND INCORRECT CUTS.

2. EXTERIOR WOOD TREATMENT CAN PROLONG THE LIFE OF YOUR PARK ELEMENTS. HOWEVER, IT IS NOT NECESSARY TO BRING MANY YEARS OF ENJOYMENT. ANY TREATMENTS MUST BE APPLIED AND MAINTAINED PROPERLY WHICH IS A LABOR INTENSIVE PROCESS. ONE SHOULD DECIDE IF THIS IS PRACTICAL OR NOT FOR THEIR PARK. THE PARK ELEMENTS ARE LOW COST, DURABLE, AND EASY TO BUILD BY DESIGN AND IT MAY BE MORE PRACTICAL TO SIMPLY HAVE THE NEXT GENERATION REBUILD THE ELEMENTS WHEN DEEMED NECESSARY. BELOW ARE RESOURCES/ LINKS FOR CONSIDERATION IF INTERESTED IN EXTERIOR WOOD TREATMENTS.

<https://www.rockler.com/learn/choosing-the-best-finish-for-outdoor-projects>
<https://www.realmilkpaint.com/blog/tips/best-finish-outdoor-wood/>
<https://vermontnaturalcoatings.com/product-category/exterior/>

BENCH + BACK \

TOOLS:

10" OR 12" MITER SAW (or appropriate hand saw)

IMPACT DRIVER for self drilling exterior wood screws

DRILL with 1/8" and 1/4" drill bit when needed for pre-drilling holes

SANDPAPER 120 GRIT

CARPENTERS SQUARE

SAFETY GOOGLES + GLOVES

MATERIALS:

(112) 2.5" SELF DRIVING EXTERIOR WOOD SCREWS

(6) 1/4" x 1 1/4" LAG EXTERIOR WOOD SCREWS

(4) 1/4" x 2 1/2" EXTERIOR CARRIAGE BOLTS AND NUTS WITH WASHER

(2) BACKREST BRACKETS -
https://www.hairpinlegs.com/products/bed-bracket-backrest?pos=1&_sid=aca35f1d7&_ss=r

LUMBER:

(7.5) 2" x 4" x 8'

1 STEP

1. MARK, LABEL AND CUT LUMBER
IN ORDER OF BB-1 TO BB-7 TO THE
FOLLOWING LENGTHS:

NOTE: CHECK EACH PIECE OF LUMBER
IMPERFECTIONS SUCH AS CRACKS, WARPING,
OR UNEVEN MILLING. DISCARD THESE
PORTIONS OF LUMBER.

BENCH + BACK ASSEMBLY

BB-1: (6) 2 x 4 x 45"
BENCH BOTTOM + TOP

BB-2: (3) 2 x 4 x 41"
BENCH BACK

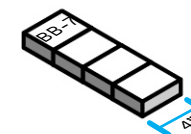
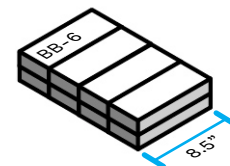
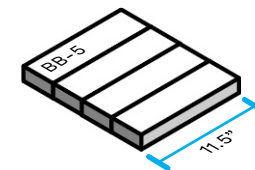
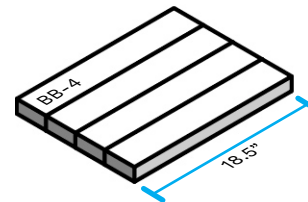
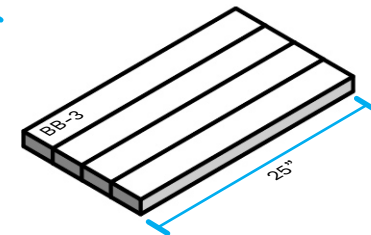
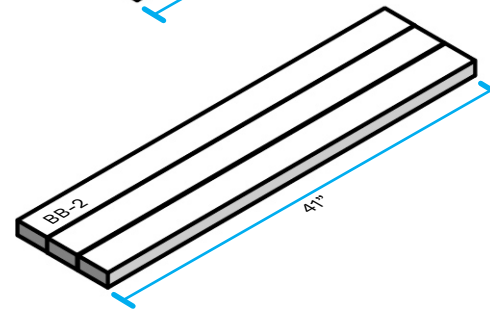
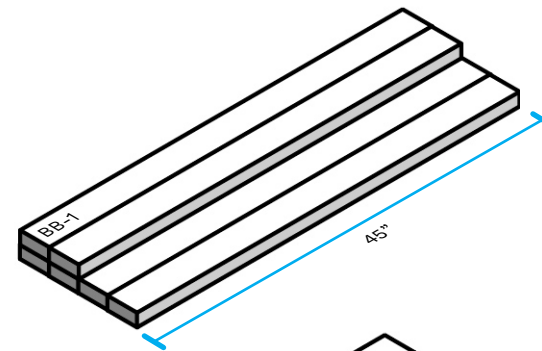
BB-3: (4) 2 x 4 x 25"
BENCH SIDE

BB-4: (4) 2 x 4 x 18.5"
BENCH SIDE

BB-5: (4) 2 x 4 x 11.5"
BENCH SIDE

BB-6: (8) 2 x 4 x 8.5"
BENCH SIDE

BB-7: (4) 2 x 4 x 4"
BENCH SIDE



TIPS:

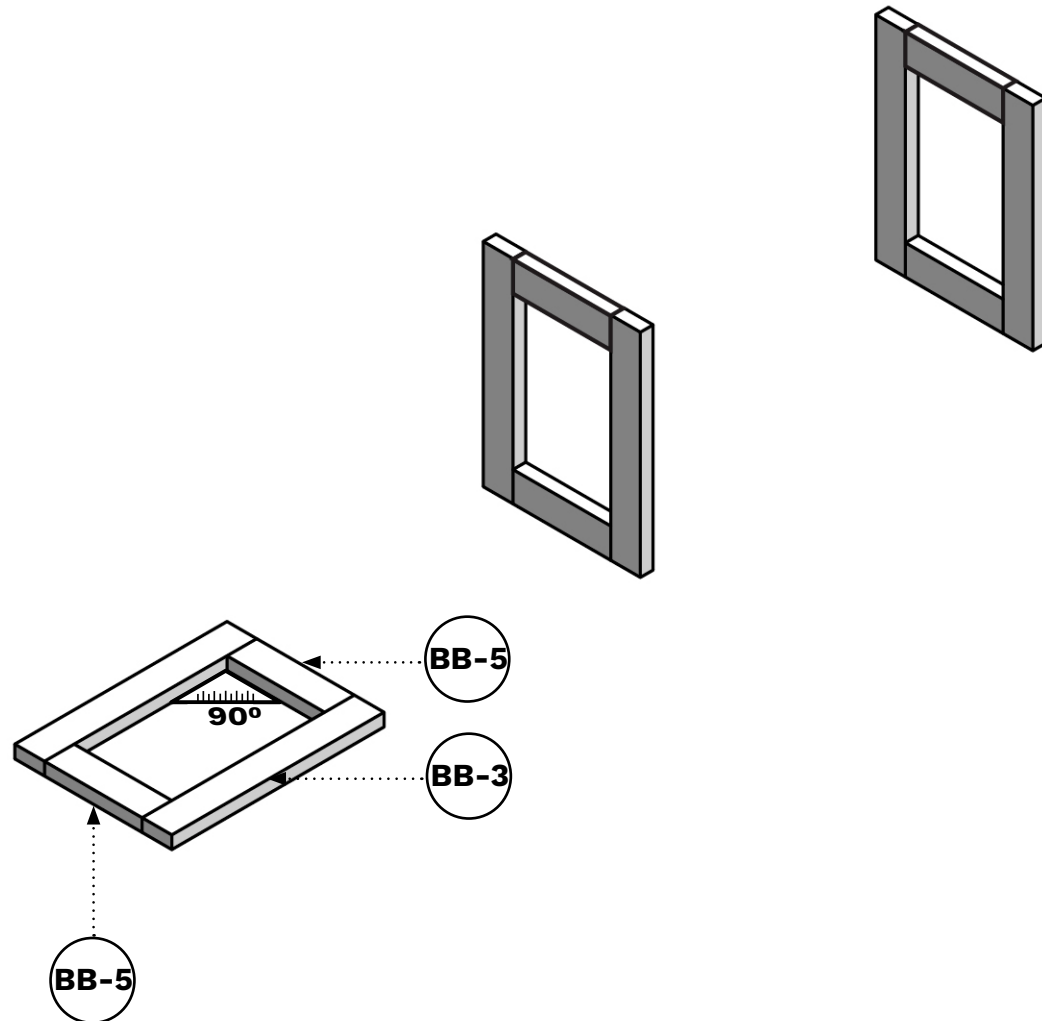
- LABEL EACH PIECE OF LUMBER AS YOU ARE MEASURING/CUTTING
- HAVE A PILE OF LUMBER FOR BB-1 TO BB-7
- LIGHTLY SANDING CUT EDGES TO REMOVE SPLINTERS WILL ALSO HELP WITH ASSEMBLY.
- IF BUILDING MULTIPLE BENCHES, CREATING A STOP FOR YOUR MITER SAW ENSURES ALL PIECES ARE CUT TO THE SAME LENGTH

2 STEP

1. PLACE (2) **BB-3** PIECES ON FLAT SURFACE.
2. PLACE (2) **BB-5** ON THE RIGHT SIDE OF THE **BB-3** PIECES, ALIGNING THE EDGES AS SHOWN IN THE IMAGE.
3. REPEAT STEPS 1-2.

NOTE: UTILIZE A CARPENTERS SQUARE TO ENSURE A RIGHT ANGLE IS FORMED IN EACH CORNER. IF THE RIGHT ANGLE IS FAR OFF, CHECK THAT YOUR MEASUREMENTS OF EACH BOARD ARE CORRECT.

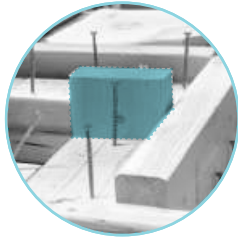
TIP: BUILDING A 90 DEGREE JIG WILL DRAMATICALLY HELP WITH ASSEMBLY OF THE NEXT TWO STEPS



3 STEP

1. PLACE (4) BB-6 AND (1) BB-4 PIECE ON TOP OF THE FRAME, AS SHOWN.

2. TO ENSURE A 1.5" GAP BETWEEN THE (2) BB-6 PIECES, USE A 2X4 PIECE OF LUMBER ON END AS A SPACER, AS SHOWN:



3. USE THE PRE-DRILLING GUIDE  AT THE END OF THE PACKET TO PRE-DRILL YOUR SCREW HOLES

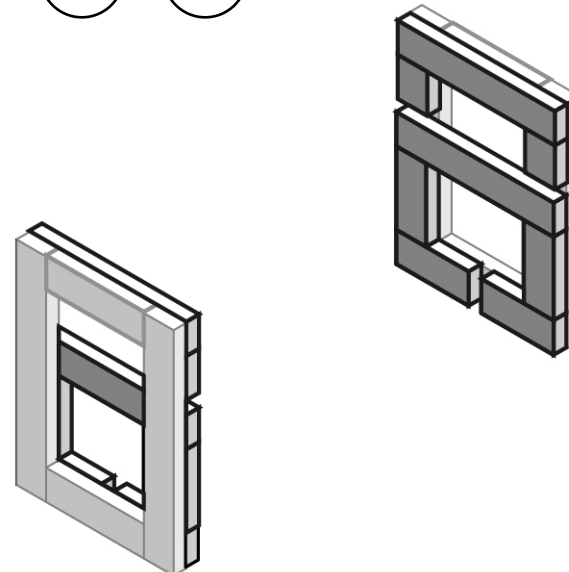
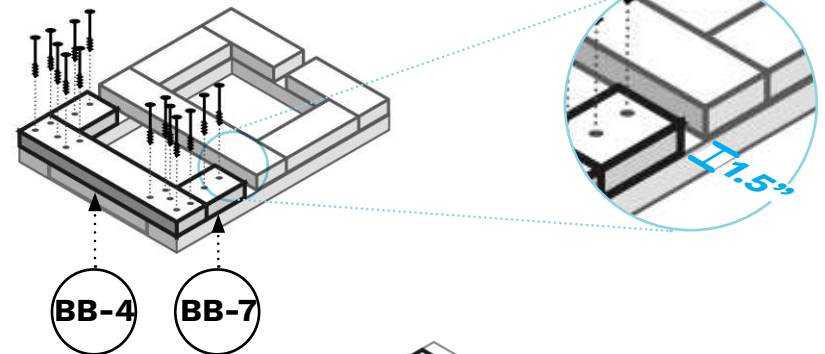
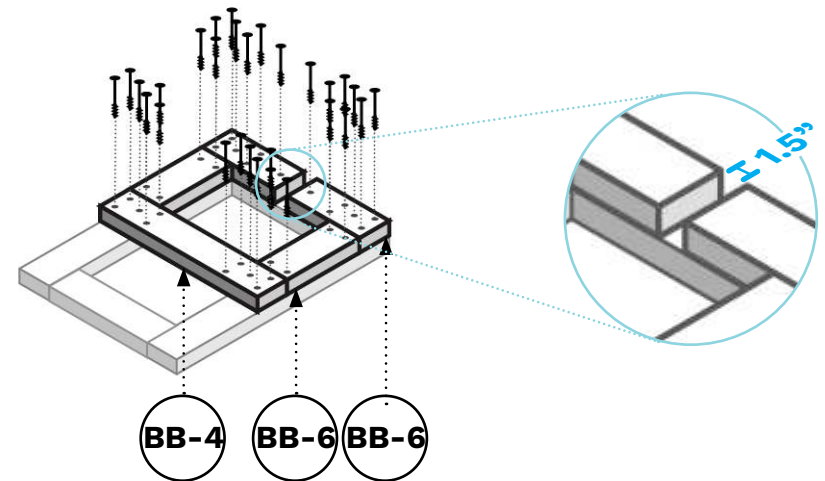
4. SECURE LUMBER BY USING AN IMPACT DRIVER WITH (30) 2.5" SELF DRIVING EXTERIOR WOOD SCREWS, AS SHOWN.

5. PLACE (1) BB-4 PIECE FLUSH TO THE BB-5 PIECE BELOW IT, AND (2) BB-7 PIECES AT RIGHT ANGLES NEXT TO IT, AS SHOWN.

6. TO ENSURE A 1.5" GAP BETWEEN THE (2) BB-7 PIECES AND THE BB-4 PIECE PREVIOUSLY SCREWED IN, USE A 2X4 PIECE OF LUMBER ON END AS A SPACER AS SHOWN IN STEP 2.

7. SECURE LUMBER BY USING AN IMPACT DRIVER WITH (14) 2.5" SELF DRIVING EXTERIOR WOOD SCREWS.

8. REPEAT STEPS 1-7 ON THE OTHER LEG, AND STAND THE LEGS UP FOR THE NEXT STEP

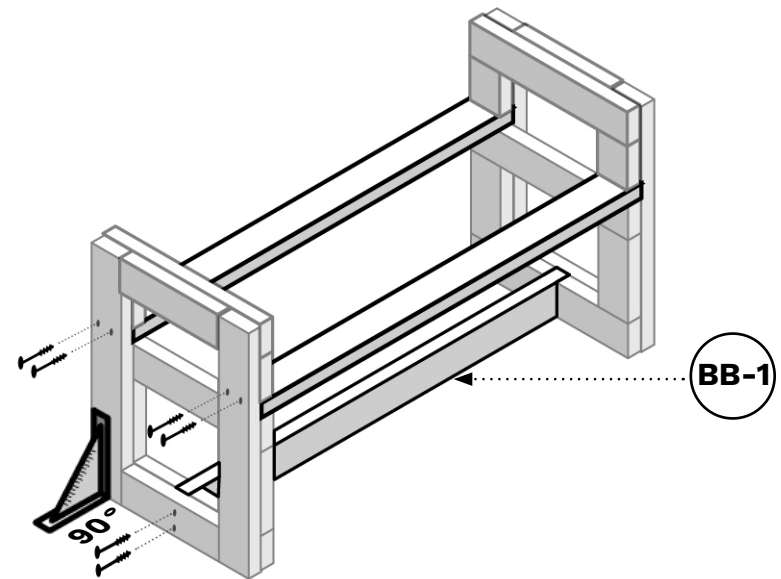


4 STEP

1. PLACE (3) BB-1 PIECES IN THE SLOTS AT THE BASE AND SIDES OF THE TWO FRAME PIECES CREATED IN STEP THREE, AS SHOWN.

TIP: ENSURE THE SIDE STAYS 90 DEGREES BY USING A CARPENTERS SQUARE

2. SECURE BB-1 LUMBER ON BOTH THE LEFT AND RIGHT SIDE INTO THE FRAME PIECES AS SHOWN, BY USING AN IMPACT DRIVER WITH (12) 2.5" SELF DRIVING EXTERIOR WOOD SCREWS TOTAL, (6) SCREWS PER SIDE.

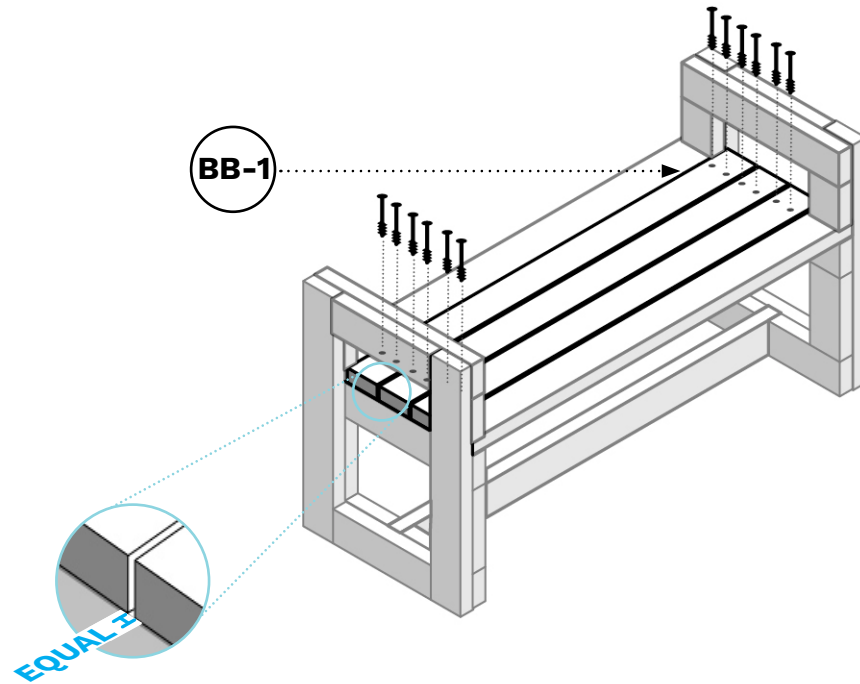


5 STEP

1. PLACE **(3)** **BB-1** PIECES ON TOP OF THE FRAME CREATED IN STEP FOUR, ALIGNING ALL EDGES AND LEAVING AN EQUAL GAP BETWEEN ALL PIECES.

2. SECURE **BB-1** LUMBER USING AN IMPACT DRIVER WITH **(12)** 2.5" SELF DRIVING EXTERIOR WOOD SCREWS.

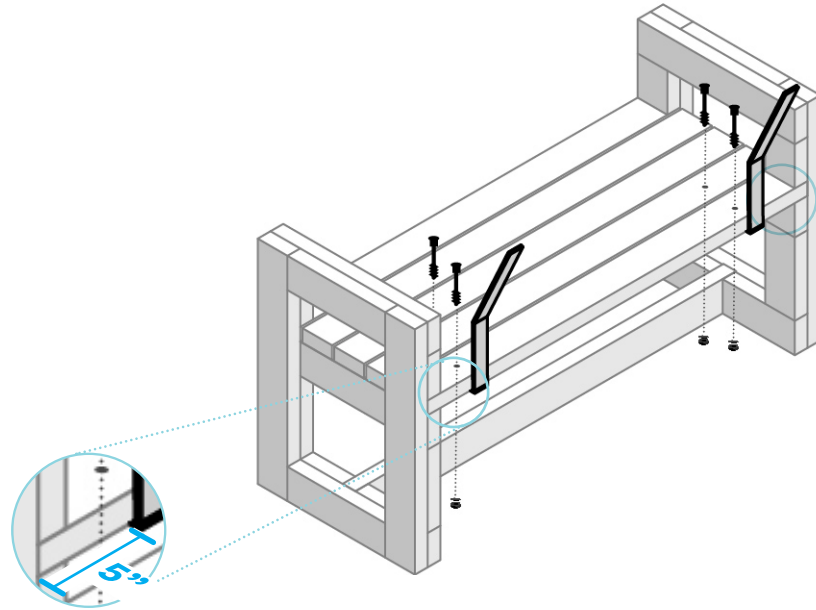
NOTE: THESE SCREWS WILL NEED TO BE DRILLED IN AT A SLIGHT ANGLE



6 STEP

TIP: TILTING THE BENCH OVER ONTO THE FRONT SIDE WILL AID IN THE COMPLETION OF THIS STEP (NOT SHOWN)

- 1. PLACE (2) BACKREST BRACKETS** ON EITHER SIDE OF THE BB-1 LUMBER **5" INWARD**, AS SHOWN.
- 2. MARK AND PRE-DRILL HOLES** TO ALIGN WITH BRACKETS STRAIGHT THROUGH BB-1 LUMBER.
- 3. SECURE BACKREST BRACKETS** INTO THE BOTTOM OF **BB-1** PIECES BY USING A SOCKET WRENCH WITH **(4) 1/4" x 2 1/2"** EXTERIOR CARRIAGE BOLTS AND NUTS WITH WASHERS.



7 STEP

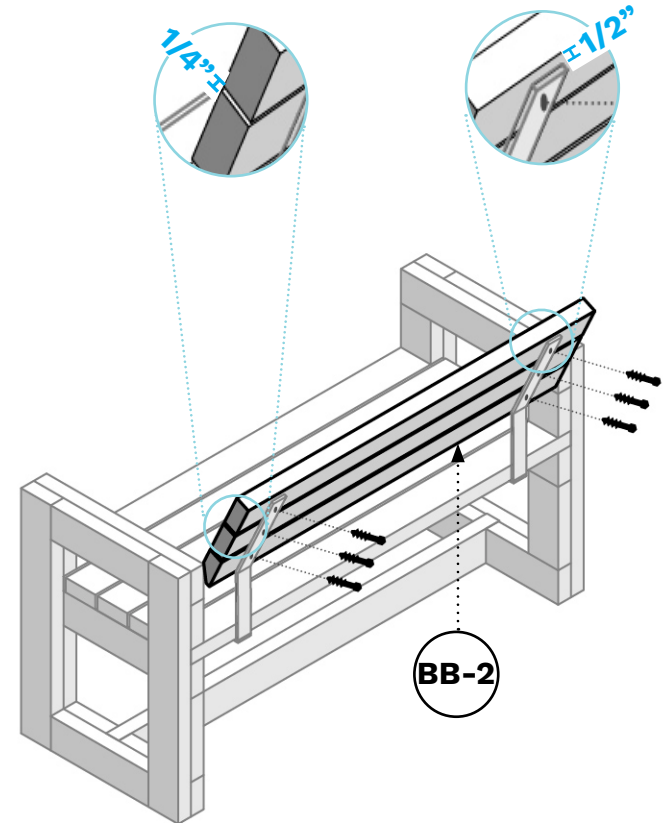
1. PLACE (3) BB-2 PIECES ON TOP OF THE BACKREST BRACKETS SECURED IN STEP SEVEN. BEGIN WITH THE TOP PIECE, SET THE PIECE 1/2" ABOVE THE TOP OF THE BRACKET. ALLOW A 1/4" GAP BETWEEN EACH PIECE BELOW IT AND ENSURE ALL EDGES ARE ALIGNED.

TIP: USE YOUR CARPENTERS SQUARE AS A SPACER.

2. SECURE BB-2 LUMBER INTO THE BACKREST BRACKETS BY USING A SOCKET WRENCH WITH (6) 1/4" x 1 1/4" LAG SCREWS.

TIP: USE 1/4" DRILL BIT TO PRE-DRILL SHALLOW HOLES TO GET LAG SCREWS STARTED. BE CAREFUL NOT TO DRILL COMPLETELY THROUGH BB-6 LUMBER.

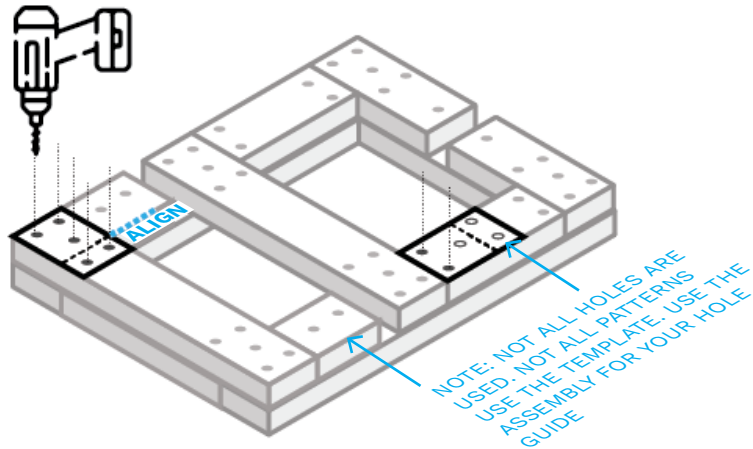
3. LIGHTLY SAND THE ROUGH EDGES OF THE BENCH + BACK.



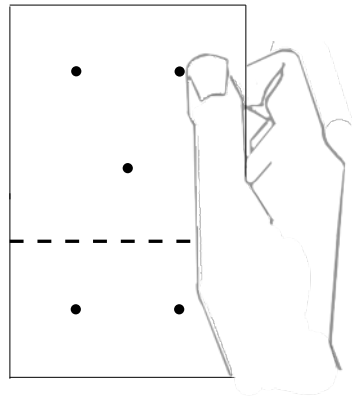


1. CUT OUT PRE-DRILLING HOLE GUIDE ALONG DOTTED LINES

2. WHEN DRILLING, ENSURE THAT YOU PROPERLY ALIGN THE EDGE OF THE TEMPLATE WITH OUTSIDE EDGE OF THE LUMBER BEFORE PRE-DRILLING



3. ENSURE ALL HOLES ARE ABOUT A THUMB IN WIDTH FROM THE EDGE (~1") EVEN HOLES THAT DON'T USE THIS TEMPLATE



PRE-DRILLING HOLE GUIDE

